1 Literature Review

This literature review is organised in three parts: a presentation of the two main families of post-hoc explanations in XAI, an overview of their use in credit scoring, and a technical review of data attribution methods for GBDTs.

1.1 Feature Attribution and Data Attribution

Two complementary families of post-hoc explanations can be distinguished in XAI. Feature attribution methods focus on understanding how a fixed model leads to particular predictions by assigning an importance score to each input feature. These include widely used techniques such as SHAP, LIME, and iBreakDown, which quantify how individual variables contribute to a model's output.

Data attribution methods address a different question: rather than explaining predictions in terms of the model, they explain where the model came from, identifying which training examples were most influential for a given prediction. In other words, they trace model behaviour back to the training data rather than to the input features. Both families can operate at the local level, explaining a single prediction, or at the global level, characterising overall model behaviour.

This conceptual distinction is important: feature attribution explains the decision logic, whereas data attribution explains the training dynamics.

1.2 Explainability in Credit Scoring

In credit scoring, explainability has become a regulatory and operational requirement, driven by the need to justify individual decisions to applicants and supervisors. In Europe, under the GDPR, Article 22(1) grants the right to individuals not to be subject to solely automated decisions with significant effect, and Article 15(1)(h) requires that, if so, they receive sufficient information about the logic involved in the decision.

The literature has largely adopted feature attribution methods to meet this need. For instance, the Quantitative Input Influence (QII) method, based on the game-theoretic concept of Shapley values, has been applied to credit default risk models to make predictions auditable for stakeholders such as lenders and applicants. Similarly, Bücker et al. (2022) provide a comprehensive framework for explainability in credit scoring, structured exclusively around feature-level methods such as SHAP, LIME, and iBreakDown. They also rely on global techniques such as permutation feature importance and partial dependence plots to assess overall model behaviour and verify whether learned relationships are economically plausible. This illustrates that data attribution is typically not framed as an explainability tool comparable to SHAP or Local Interpretable Model-agnostic Explanations (LIME) methods.

Additional studies reinforce this trend. Provenzano et al. (2020), for example, propose a machine-learning framework for credit scoring that integrates heterogeneous data sources and applies probability calibration to ensure that predicted default probabilities align with observed frequencies. They then map these calibrated probabilities into internal rating grades using a genetic algorithm with monotonicity and cluster-quality constraints. Explainability is again addressed through SHAP and LIME, used to provide local explanations of individual credit decisions.

However, the reliability of these feature attribution methods has been questioned. Ratul et al. (2021) show that both SHAP and LIME exhibit important limitations in practice: features identified as important for one class often appear as important for the opposite class as well, explanations vary significantly across similar instances within the same class, and the two methods show substantial disagreement in their feature rankings. This suggests that feature attribution alone may not provide stable or representative explanations, further motivating the use of complementary approaches.

Beyond these methodological concerns, recent work in fairness auditing shows that discriminatory outcomes can often be traced back to the composition of the training data rather than to model design alone. Their finding directly motivates the use of methods able to identify which specific training applicants shaped a given prediction, a question that feature attribution cannot answer. The scarcity of data attribution applications in credit scoring can potentially be explained by the late adaptation of influence functions to GBDTs, only made computationally tractable on tabular pipelines by recent work.

1.3 Evolution of Data Attribution Methods for GBDTs

The naive approach to data attribution, Leave-One-Out (LOO) retraining, measures the influence of a training instance z_i on a target example z_e as the change in loss between the model trained on the full dataset and the model trained without z_i. This process requires retraining the model for each removed point, making it computationally prohibitive. Fortunately, LOO can be approximated via the influence function by computing the parameter change through an infinitesimal upweighting of a data point and an inverse Hessian computation. ? adapted this framework to modern deep learning models, in particular to neural networks. However, their method requires the model to be differentiable, which excludes GBDTs.

Sharchilev et al. (2018) extended influence estimation to GBDTs under a fixed-tree-structure assumption: removing a training instance is assumed not to alter the splits learned by the trees, so that only leaf values need to be recomputed. Under this assumption, they introduced LeafRefit, which recomputes all leaf values without z_i and is more efficient than full LOO, and LeafInfluence, which further approximates leaf-value perturbations analytically via the influence function frame- work of ?. However, because GBDTs are trained sequentially, perturbing leaf values in one tree alters residuals in subsequent trees, making LeafInfluence paradoxically slower than LeafRefit and even full retraining.

Both LeafRefit and LeafInfluence estimate influence by comparing the model's final state with and without the effect of z_i, in a counterfactual logic. BoostIn, developed by ?, takes a fundamentally different perspective by adapting the TracIn method of Pruthi et al. (2020) to GBDTs. Rather than comparing two endpoints, BoostIn traces the influence of z_i on z_e throughout the entire training process: at each boosting iteration, it measures how much z_i contributed to reducing the loss on z_e via a dot product of their respective gradients, restricted to iterations where both points share the same leaf. The total influence is the sum of these contributions across all trees. By avoiding the propagation of perturbations through the ensemble, BoostIn achieves substantially lower computational cost than LeafInfluence.

Source à ajouter:

- Provenzano, A. R., Trifiro, D., Datteo, A., Giada, L., Jean, N., Riciputi, A., ... & Nordio, C. (2020). Machine learning approach for credit scoring. *arXiv preprint arXiv:2008.01687*.

- Ratul, Q. E. A., Serra, E., & Cuzzocrea, A. (2021, December). Evaluating attribution methods in machine learning interpretability. In *2021 IEEE International Conference on Big Data (Big Data)* (pp. 5239-5245). IEEE.
- Bücker, M., Szepannek, G., Gosiewska, A., & Biecek, P. (2022). Transparency, auditability, and explainability of machine learning models in credit scoring. *Journal of the Operational Research Society*, *73*(1), 70-90.